

A Comprehensive Investigation on Comparison of Machine Learning Techniques on Heart Disease Domain

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Abstract—This paper aims to investigate and compare the accuracy of different data mining classification schemas and their combinations under Ensemble Machine Learning Techniques for predicting heart disease. The Cleveland data set for heart diseases containing 303 instances is used as the main database for training and testing of the developed system. Due to the limited number of data, 10-Fold Cross-Validation is applied in order to boost it. In this study different classifiers, such as Decision Tree (DT), Nave Bayes (NB), Multilayer Perceptron (MLP), K-Nearest Neighbor (K-NN), Single Conjunctive Rule Learner (SCRL), Radial Basis Function (RBF) and Support Vector Machine (SVM) are utilized. Moreover, the ensemble prediction of classifiers, bagging, boosting and stacking, are applied on the dataset. The result of the experiment indicates that SVM method using boosting technique outperformed among the aforementioned methods.

I. INTRODUCTION

Coronary illness is the most noteworthy reason for death on the planet in the course of recent years [?]. Recently, heart disease has become one of the most prevalent diseases that people are being suffered. In other words, heart disease is the most significant cause of death in the world over the past 10 years [?]. Many factors such as clinical symptoms and the relation between the functional and the pathologic manifestations of heart diseases and other human organs rather than heart, complicate the diagnosis of it and result in delay in correct diagnosis decision. According to [?], clinical decisions are often made based on doctors intuitions and heuristics experience rather than on the knowledge rich data hidden in the database. Therefore, it leads to unwanted biases, errors and excessive medical costs which affect the quality of treatment provided to patients [?].

Based on the high percentage of mortality through heart disease and availability of vast amount of patients records in hospitals data base, diagnosing of heart disease is an essential matter in health care industry and many researchers try to find the most accurate machine learning techniques to discover the relationship between different heart disease and patient attributes to help physicians [?], [?].

This paper relies on a comparison of different machine learning classification techniques such as Decision Tree (DT),

Nave Bayes (NB), MultiLayer Perceptron (MLP), K Nearest Neighbor (K-NN), Single Conjunctive Rule Learner (SCRL), Radial Basis Function (RBF), Support Vector Machine (SVM), and combination of them on heart disease data set using three main data portioning models: 10-Fold Cross Validation. The dataset used is the Cleveland Heart Disease data set taken from UCI learning data set repository which was donated by Detrano. The data set is being divided into two classes, “False” corresponding to absence of heart disease and “True” representing the presence of disease.

The rest of the paper is organized as follows. Related works are presented in section II. In section III, research methodology is described. In section IV, experimental results and analysis are presented and discussed in details. In section V, comparison of data mining techniques is discussed. The study is finally concluded in section VI.

II. RELATED WORKS

In the last few years, several studies have been dedicated to the evaluation of the classification accuracies of several classification algorithms applied to the Cleveland heart disease database [?] freely available at an online data mining repository of the University of California, Irvine (UCI).

This database contains 4 databases concerning heart disease diagnosis. In particular, it contains the dataset collected by Robert Detrano in 1988 at the V.A. Medical Center, Long Beach and Cleveland Clinic Foundation. This dataset is composed by 303 instances which makes it small resource.

Since its creation, it was used from a lot of researchers in different classification problems face with various classification algorithms. Same Robert Detrano, who constructed the Cleveland heart disease database, used a logistic regression algorithm and obtained 77.0% classification accuracy [?].

In 1995, papers like [?] investigated machine learning techniques, as a Flexible Bayes and/or a Naive Bayes network, achieving 84.66% of accuracy.

In [?], the author worked on the Cleveland data set with focus on comparing global evolutionary computation

approaches, and observed some prediction performance improvement with a new approach. However, performance of the proposed technique is dependent on the attributes selected by the algorithm.

Gudadhe et al. [?] realized a architecture base both on multilayer perceptron neural network (MLP) and a Support Vector Machine (SVM) approach. This architecture achieved an accuracy of 80.41% in term of classification between two classes (presence of heart disease or absence of heart disease).

On the other hand, Humar Kahramanli and Novruz Allahverdi [?] obtained an accuracy of 87.4% by using a hybrid neural network that combines a fuzzy neural network (FNN) with an artificial neural network (ANN).

In [?] a neural networks ensemble method was included in a methodology which uses SAS Base Software 9.1.3 for diagnosing of the heart disease. The classification method based on Support Vector Machine(SVM) and integer-coded genetic algorithm (GA) were used for heart disease detection and it obtained 89.01% classification accuracy. The same result was obtained from Das et al. [?].

In [?] an Intelligent Heart Disease Prediction System (IHDPs) is presented. The IHDPs system uses data mining techniques, as for example Decision Trees, Nave Bayes and Neural Network. The tests performed by the authors indicated that Nave Bayes model achieved the best performance in terms of correct predictions (86.12%). The second model was the Neural Network with a correct predictions of 86.12% and finally the Decision Trees with a score of 80.4% of correct predictions.

III. RESEARCH METHODOLOGY

Important concepts such as data set, data portioning model and data mining techniques are described in this section.

A. The Cleveland Data Set

As it was mentioned before, the Cleveland data set is used in this paper. This data set contains 303 instances with total number of 76 attributes, however majority of the studies use a maximum of 14 attributes [?] as these are considerably linked to the heart disease [?]. These 14 attributes are represented in table I.

1) *Data Portioning Technique:* For selecting training set and testing set and due to data limitation, 10-Fold Cross Validation, is used. The main reason for choosing 10-fold cross validation is that it has lower variance in comparison with other estimators like single hold out. In single hold out set, 90% of data is used for training set and 10% for testing set; thus, the test case is too small and will lead to performance variation for different partition of data. 10-fold cross validation reduces this variance by averaging over k different partitions. In this case, the performance is less sensitive to the partitioning of data. In the following section K-Fold Cross validation is explained in detail.

TABLE I
DESCRIPTION OF ATTRIBUTES OF CLEVELAND DATASET

Attribute	Description (unit)
Age	Age expressed in year (int)
Sex	Gender (int): - male = "0" - female = "1"
CP	Chest pain type (int): - Typical Angina = "1" - Atypical Angina = "2" - Non-Angina = "3" - Asymptomatic = "4"
Trestbps	Resting Blood Pressure (mmHg)
Chol	Serum Cholesterol (mg/dl)
Fbs	Index about fasting glucose (int): - fasting glucose $\leq$ 120 mg/dl = "0" - fasting glucose $>$ 120 mg/dl = "1"
Restecg	Results about the Resting ECG (int): - Normal = "0" - ST-T wave abnormality = "1" - Probable or definite left ventricular hypertrophy = "2"
Thalach	Maximum heart rate achieved (beats/s)
Exang	Index about exercise induced angina - No = "0" - Yes = "1"
Oldpeak	Index about exercise induced ST depression at rest - No = "0" - Yes = "1"
Slope	The slope of the peak exercise ST segment - Up sloping = "1" - Flat = "2" - Down sloping = "3"
Ca	Number of major vessels coloured by fluoroscopy (int)
Thal	Summary of heart condition - Normal = "3" - Fixed Defect = "6" - Reversible Defect = "7"
Num	Class attribute (boolean): - Existence of heart disease = "true" - Healthy = "false"

2) *K-Fold Cross Validation:* Among the data portioning techniques, K-Fold Cross validation is the recommended one when we are dealing with limited amount of data for training and testing sets. In this method, in each iteration, based on the number of folds (K), the dataset is divided into two parts for training and testing. For example, in 3-Fold cross validation, data is split into three equal partitions each of them in turn is used for testing and the reminder for training. Extensive tests on numerous different dataset with different techniques have shown that 10 are about the right number of fold to get the best estimate of error [?].

B. Machine Learning Techniques

The main goal of current study is comparing the different machine learning techniques on a small dataset. In addition, we try to improve the accuracy of method to have a better comparison in our experiment. For this purpose, 7 classifiers such as, DT, NB, MLP, RFB, SCRL, K-NN, SVM and combination of them through ensemble learning methods such as, bagging, boosting and stacking are discussed. In each scenario, the performance is calculated using the standard metrics such as, accuracy, precision, recall and F-measure, which are evaluated using predictive classification table, known as Confusion Matrix. In addition, Receiver Operation Characteristic (ROC) curve area is utilized to compare the performance of each classifier.

1) *Decision Tree (DT)*: A Decision Tree is a flowchart-like structure that includes a root node, branches, and leaf nodes. Dataset attributes is defined through the internal nodes. The branches are the outcome of each test against each node. It is a popular classifier because it is simple, fast, and easy to interpret, explain and implement. It requires no domain knowledge or parameter setting.

2) *Nave Bayes (NB)*: Nave Bayes classifier is based on Bayes' Theorem. In this classifier independency between attributes of the data set is the main assumption and the key point in order to make a prediction. It is easy to implement and particularly useful for very large data sets. In addition to simplicity, this model is shown to outperform even highly sophisticated classification methods. One of the advantages of Nave Bayes classifier is that it requires small amount of data for parameters (e.g., means and variance) estimation for classification.

3) *Multilayer Perceptron (MLP)*: Multilayer Perceptron (MLP) classifier contains multiple of nodes arranged in layers. This makes a directed graph which covers input, hidden, and output layers and each layer is fully connected to the next one. This classifier has a clear architecture and simple algorithm; therefore, it is one of the famous neural network models.

For this experiment, we tried combination of different numbers of input, hidden and output and finally came up with the best combination, which are 13 inputs, 7 hidden layer and 2 outputs. This combination let us get better results.

4) *Support Vector Machine (SVM)*: Support Vector Machine models are defined a finite-dimensional vector space that each dimension represents a "feature" of a particular object. It has shown to be an effective approach in high-dimensional space problems. Due to its computational efficiency on large datasets this technique is usually used in document classification and sentiment analysis

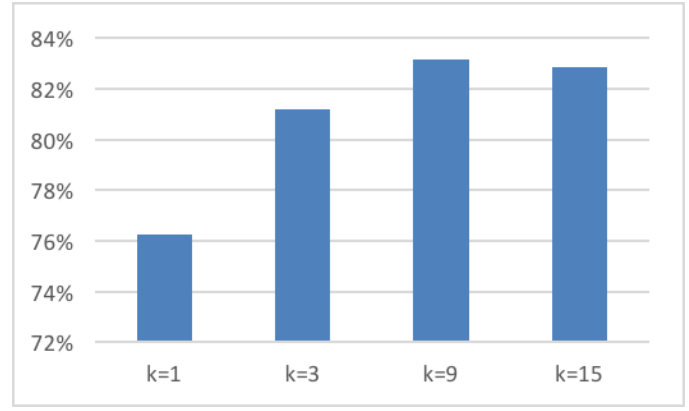


Fig. 1. Accuracy of K-NN based on different value of K.

5) *K-Nearest Neighbors (K-NN)*: K-Nearest Neighbors (K-NN) is an instance-based learning or lazy learning, which generalizes beyond the training data is postponed until testing time. When a query is made, the training data will be generalized. K-NN classifies an object by majority vote of its closest neighbors. In other words, based on some distance metrics, the class of a new instance will be predicted. The distance metric used in nearest neighbor methods for numerical attributes can be simple Euclidean distance. K can have different odd values such as 1, 2, 3 and so forth.

For this experiment, we tried for K=1, 3, 9 and 15 and the best value for K was 9. The accuracy of K-NN with different numbers of K is shown in the below figure. As figure 1 shows, K=9 has the most accuracy (84%) and when we continued the experiment the accuracy was degraded.

6) *Radial Basis Function (RBF)*: Similar to MLP, RBF is also another popular classifier based on Artificial Neural Network(ANN). RBF networks derive from the theory of function approximation and uses a static Gaussian function as the nonlinearity for the hidden layer processing elements. Both MLP and RBF networks are nonlinear layered networks which are using universal approximation properties. In comparison with MLP, RBF networks take a slightly different approach in terms of number of hidden layers, linearity of output layer, local and global approximations, and number of parameters. The main advantage of this model is selecting suitable hidden unit without a full nonlinear optimization of the whole network.

7) *Single Conjunctive Rule Learner (SCRL)*: Single conjunctive rule learner or inductive learning is one of the machine learning algorithms that aims to infer a set of rules from dataset that captures all generalizable knowledge within that data. Rules in this model can be various and usually are in order from more general to more specific. During the classification process, the first rule that matches with the new instance will stop the classification.

TABLE II
STANDARD METRICS FOR 10-FOLD CROSS VALIDATION TECHNIQUE

Model	TP Rate	FP Rate	Precision	Recall	F-Measure	ROC Area
DT	0.83	0.29	0.774	0.83	0.801	0.8
NB	0.867	0.203	0.836	0.867	0.851	0.904
k-NN(k=1)	0.782	0.261	0.782	0.782	0.782	0.752
k-NN(k=3)	0.836	0.217	0.821	0.836	0.829	0.838
k-NN(k=9)	0.842	0.181	0.848	0.842	0.845	0.898
k-NN(k=15)	0.836	0.181	0.847	0.836	0.841	0.904
MLP	0.824	0.21	0.824	0.824	0.824	0.894
RBF	0.861	0.188	0.845	0.861	0.853	0.892
SVM	0.897	0.225	0.827	0.897	0.86	0.836
SCRL	0.703	0.304	0.734	0.703	0.718	0.707

8) *Ensemble Learning Methods*: This is aggregation of prediction of several classifiers. It has three different models: bagging, boosting and stacking [14]. Bagging is combining predictions of exactly the same types by voting. Boosting is the same as bagging while the performance of previously models effect on new models. Stacking is combining models of different types.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

The result of applying seven aforementioned methods (DT, NB, K-NN, SVM, MLP, RBF, SCRL and combination of them) using 10-Fold Cross Validation are presented in this section.

The main performance measurements used in this study are Precision, Recall, F-Measure and Receiver Operating Characteristics (ROC) graph . Precision (also called positive predictive value) is a measure of result relevancy, while recall (also known as sensitivity) is a measure of how many truly relevant results are returned. In a classification, a higher precision percentage is related to a low false positive rate, and a higher recall percentage is correlated to a low false negative rate. A classifier with a higher recall but lower precision will show many results, but most of the predictions are labeled incorrectly. On the other hand, a classifier with a higher precision and a lower recall will depict very few results, but most of the predictions are labeled correctly. As both measures, precision and recall, are important, we usually evaluate our classifier with F-measure which is the combination of precision and recall. ROC graphs is another technique to examine and visualize the performance of a classifier. ROC graph shows the tradeoff between the ability of a classifier to correctly identify positive cases and the number of negative cases that are incorrectly classified.

1) *Result of Experiment through 10-Fold Cross Validation*: In the first experiment we use the whole dataset and apply DT, NB, MLP, SVM, K-NN, RBF and SCRL approaches while using 10 fold cross validation technique to estimate the performance of each method. For K-NN classifier, we tried for K=1,3,9 and 15 and the best value for K was 9. The

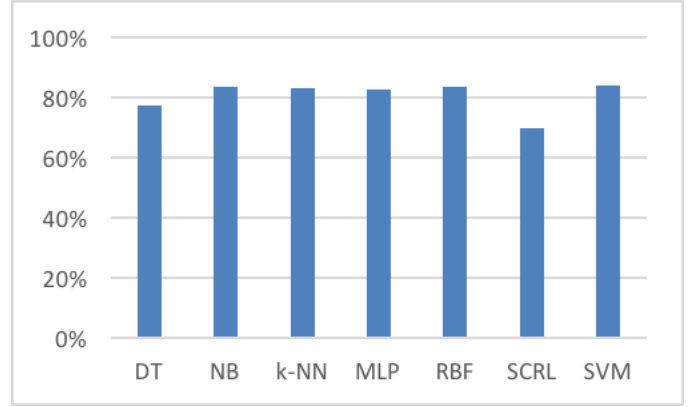


Fig. 2. The performance of seven methods through 10-fold Cross Validation technique.

accuracy of K-NN with different numbers of K is depicted in the figure 1. As figure 1 shows, K=9 has the most accuracy (84%) and when we continued the experiment the accuracy was degraded. Also, for MLP method, we tried combination of different numbers of input, hidden and output and finally came up with the best combination, which are 13 inputs, 7 hidden layer and 2 outputs. This combination let us get better results.

As figure 2 shows, Decision tree has 77.55% accuracy, Nave Bayes 83.49%, K-NN 83.16%, Multilayer Perceptron with 82.83%, RBF with 83.82%, SCRL 69.96% and SVM 84.16%. Among them, SVM reach the highest percentage of accuracy while RBF, Nave Bayes and K-NN are in the next place. Among all single classifiers, SCRL has the lowest accuracy of 69.96%. Table ?? shows more details about standard metrics of accuracy for this experiment.

A. Result of Experiment from Ensemble Learning Methods

1) *Bagging*: Based on the experiment results, we explored the ability of Bagging techniques on improving the performance of estimation. As figure 3 shows, bagging improved some classifier accuracy. For instance, decision tree improve to

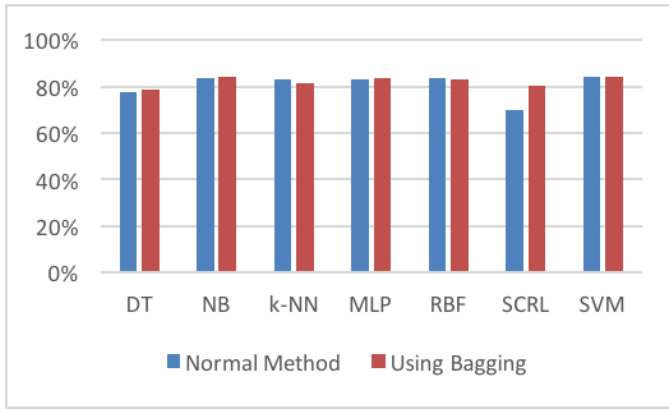


Fig. 3. Accuracy after applying Bagging technique.

78.54% from 77.55%, SCRL accuracy increased from 69.96% to 80.52% and SVM remained the same. The most significant effect was on SCRL. However, SVM maintained its accuracy.

2) *Boosting*: Boosting means applying weak classifier and run it multiple times on training data and then let the learned classifier votes. In this experiment, we applied boosting on DT, NB, K-NN, MLP, RBF, SCRL and SVM and as we expected it worked well on weakest classifiers. It changed the accuracy of DT from 77.55% to 82.17% and SCRL from 69.96% to 81.18%. The others remained almost the same.

3) *Stacking*: In this section combination of different methods for stacking is shown in the below table. As the table III shows, combination of SVM and MLP has the best accuracy which is 84%.

TABLE III
ACCURACY OF DIFFERENT METHODS THROUGH STACKING TECHNIQUE

Stacking	Accuracy
NB, K-NN, MP, RBF, SVM	78.54
NB, MP, RBF, SVM	82.17
NB, MP, RBF, SVM, SCRL	82.17
SMO, MLP	82.50
SVM, MLP	84.15
SVM, MLP, DT	83.16

V. COMPARISON OF MACHINE LEARNING TECHNIQUES

The result of four experiments shows that SVM and SCRL with 84.15% and 69.96% have highest and lowest accuracy among the others. After applying Bagging still SVM with the same accuracy percentage is the best approach while DT with 78.54% is the worst one. When we use Boosting technique SVM wins again with 84.81%. The stacking technique also confirms that the combination of SVM and MLP has highest accuracy with 84.15%. ROC curves of three most accurate prediction algorithms are shown in figure 4, 5, and 6. These confirms that SVM is the best technique in our study.

VI. CONCLUSION

In this paper, different machine learning techniques including Decision Tree (DT), Nave Bayes (NB), Multilayer

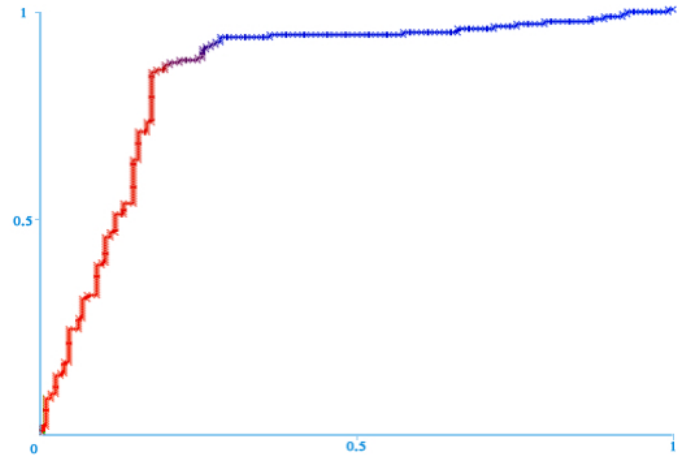


Fig. 4. ROC curve Stacking SVM, MLP, DT(Area under curve:0.8456).

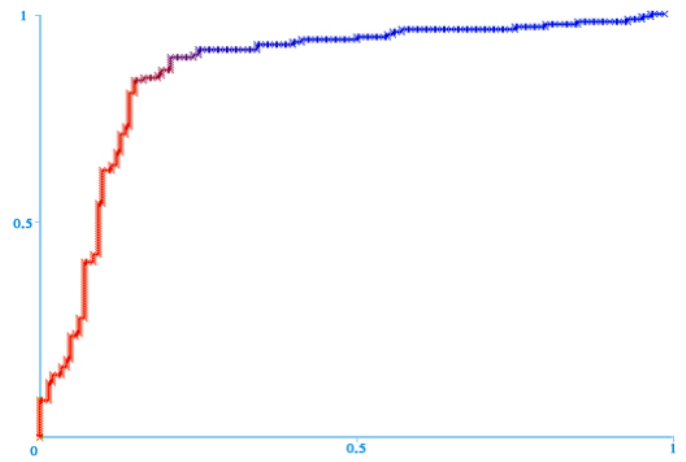


Fig. 5. ROC curve Stacking SVM, MLP (Area under curve: 0.8619).

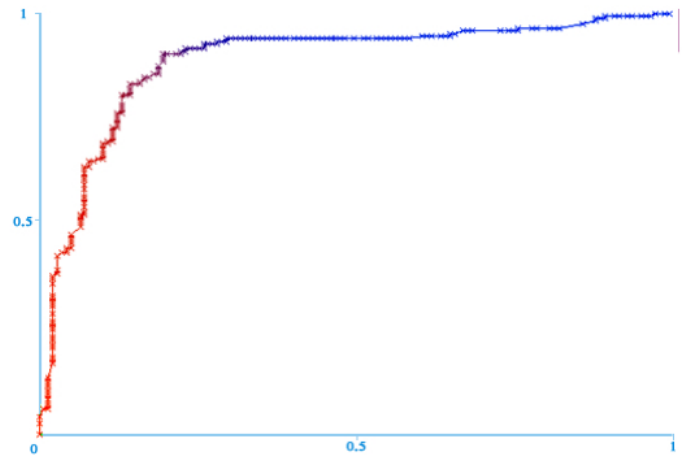


Fig. 6. ROC curve Boosting SVM (Area under curve: 0.8846).

Perceptron (MLP), K-Nearest Neighbor (K-NN), Single Con-junctive Rule Learner (SCRL), Radial Basis Function (RBF) and Support Vector Machine (SVM) and their combinations

through ensemble machine learning approaches were applied on Cleveland Heart Disease data set in order to compare the performance of each method. As the number of instances in this experiment is limited to 303 instances, the 10-Fold Cross Validation for portioning the data to training and testing datasets is selected. Each technique was run with different parameters to receive higher accuracy before comparison. The efficiency of each single classifier and also the combination of them through bagging, boosting, and stacking is evaluated. In general, we had some improvements (from significantly to almost nothing) in each technique after applying bagging, boosting, and stacking approaches and from seven different techniques, SVM outperformed among the all via boosting method. Although, our goal in this paper was to compare different machine learning techniques on a small dataset but we tried to improve the accuracy of aforementioned techniques to have a better comparison.

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